1. Graph Representation
The road network is represented as a directed graph.

Nodes (|V| = 4): road junctions.
Edges (|E| = 5): directed road segments.

Node features ($d\_v = 6$) describe junction properties.
Edge features ($d\_e = 5$) describe road characteristics.

- N = 4 junctions (nodes)
- |E| = 5 directed road links (edges)
- d_v = 6 node features
- d_e = 5 edge features

2. Node Features
Each junction has a 6-dimensional feature vector:

φ_flow – historical traffic flow
φ_signal – signal cycle time
φ_type – junction type
φ_x – x coordinate
φ_y – y coordinate
φ_deg – number of connecting roads

Node Feature Vector

For junction v_1 (a signalized intersection in the city center):

Index	**Feature**	**Symbol**	**Value**	**Units**
0	Average historical flow	φ_flow	1200	veh/hr
1	Signal cycle time	φ_signal	90	seconds
2	Junction type	φ_type	1	(signalized)
3	X coordinate	φ_x	0.35	normalized
4	Y coordinate	φ_y	0.72	normalized
5	Connecting roads	φ_deg	4	count

$\mathbf{x_1} = [1200, 90, 1, 0.35, 0.72, 4]^T \in \mathbb{R}^6$

Complete node feature matrix $X \in \mathbb{R}^{4x6}$:

	φ_flow	**φ_signal**	**φ_type**	**φ_x**	**φ_y**	**φ_deg**
v_1	1200	90	1	0.35	0.72	4
v_2	800	60	1	0.65	0.80	3
v_3	400	0	0	0.50	0.40	3
v_4	1500	120	2	0.80	0.25	4

3. Edge Features
Each road segment has a 5-dimensional feature vector:

ε_cap – road capacity
ε_speed – speed limit
ε_lanes – number of lanes
ε_len – link length
ε_type – road type

Edge Feature Vector

For edge e_{12} (arterial road from v_1 to v_2):

Index	**Feature**	**Symbol**	**Value**	**Units**
0	Road capacity	ε_cap	1800	veh/hr
1	Speed limit	ε_speed	50	km/h
2	Number of lanes	ε_lanes	3	count
3	Link length	ε_len	1.2	km
4	Road type	ε_type	1	(arterial)

Raw edge feature vector:
$\mathbf{e}_{12}$ = [1800, 50, 3, 1.2, 1]^T $\in \mathbb{R}^5$

4. Feature Normalization
Features are normalized using z-score normalization:

$\hat{x} = (x - \mu) / \sigma$

Values: [1200, 800, 400, 1500]
Normalized value for v_1:
$\hat{x}_1[0] = (1200 - 975) / (414.7 + 10^{-8}) = 225 / 414.7 \approx 0.543$
After normalizing all features:
$\hat{x}_1 \approx [0.543, 0.267, 0.577, -0.816, 1.155, 0.577]^T$

5. Classical Node Embedding
Node features are projected into an 8-dimensional hidden space.

Classical Embedding Layer
W_v $\in \mathbb{R}^{8\mathbf{x}6}$

```
W_v = [[ 0.2, -0.1,  0.3,  0.1,  0.0, -0.2],
       [-0.3,  0.4,  0.1, -0.2,  0.3,  0.1],
       [ 0.1,  0.2, -0.4,  0.3,  0.1,  0.0],
       [ 0.0, -0.3,  0.2,  0.1,  0.4, -0.1],
       [ 0.4,  0.1,  0.0, -0.1,  0.2,  0.3],
       [-0.2,  0.0,  0.3,  0.2, -0.3,  0.1],
       [ 0.1,  0.3, -0.1,  0.4,  0.0,  0.2],
       [ 0.3, -0.2,  0.2, -0.3,  0.1,  0.4]]
```

b_v = [0.1, 0.0, -0.1, 0.05, 0.0, 0.1, -0.05, 0.0]

Calculation for node v_1:

z = W_v · $\hat{\mathbf{x}}_1$ + b_v

z[0] = 0.2(0.543) + (-0.1)(0.267) + 0.3(0.577) + 0.1(-0.816) + 0.0(1.155) + (-0.2)(0.577) + 0.1
= 0.109 - 0.027 + 0.173 - 0.082 + 0 - 0.115 + 0.1
= 0.158

After computing all 8 dimensions and applying ReLU:

$\mathbf{h}_1^{(0)} = \mathrm{ReLU}(z) = [0.158, 0.0, 0.247, 0.0, 0.412, 0.089, 0.331, 0.0]^T$

6. Message Passing
Nodes exchange information through edges.

Message Construction

Consider computing the message from v_2 to v_1 (for edge e_{21}, assuming it exists):

Given:

$\mathbf{h}_2^{(0)} = [0.231, 0.0, 0.189, 0.145, 0.0, 0.267, 0.0, 0.312]^T \in \mathbb{R}^8$

$\mathbf{f}_{21} = [0.156, 0.423, -0.211, 0.089, 0.577]^T \in \mathbb{R}^5$ **(embedded edge features)**

Step 1: Concatenation

$[\mathbf{h}_2^{(0)} \parallel \mathbf{f}_{21}] = [0.231, 0.0, 0.189, 0.145, 0.0, 0.267, 0.0, 0.312, 0.156, 0.423, -0.211, 0.089, 0.577]^T \in \mathbb{R}^{13}$

Step 2: First MLP Layer

$\mathbf{W}_1^{\mathbf{m}} \in \mathbb{R}$^(2d_h × 2d_h) = $\mathbb{R}$^(16 × 16)

hidden = ReLU($\mathbf{W}_1^{\mathbf{m}} \cdot [\mathbf{h}_2 \parallel \mathbf{f}_{21}] + \mathbf{b}_1^{\mathbf{m}}$)

Step 3: Second MLP Layer

$\mathbf{W}_2^{\mathbf{m}} \in \mathbb{R}$^(d_h × 2d_h) = $\mathbb{R}$^(8 × 16)

$\mathbf{m}_{2\to1}^{(1)} = \mathbf{W}_2^{\mathbf{m}} \cdot \mathbf{hidden} + \mathbf{b}_2^{\mathbf{m}} \in \mathbb{R}^8$

Example output:

$\mathbf{m}_{2\to1}^{(1)} = [0.145, 0.267, 0.0, 0.089, 0.312, 0.178, 0.0, 0.234]^T$

Neighborhood Aggregation

For junction v_3 with incoming edges from v_1 and v_2:

N_(v_3) = {v_1, v_2}

$\mathbf{m}_{1\to3} = [0.2, 0.1, 0.3, 0.0, 0.15, 0.25, 0.1, 0.0]^T$
$\mathbf{m}_{2\to3} = [0.1, 0.3, 0.0, 0.2, 0.1, 0.0, 0.2, 0.15]^T$

7. Neighborhood Aggregation
Messages from neighboring nodes are aggregated.

Aggregation (sum):

$$\mathbf{a}_3^{(1)} = \mathbf{m}_{1\to3} + \mathbf{m}_{2\to3} = [0.3, 0.4, 0.3, 0.2, 0.25, 0.25, 0.3, 0.15]^T$$

8. Residual Update
The aggregated message is added to the previous node embedding.

Residual combination:

$$\mathbf{h}_3^{(0)} = [0.15, 0.0, 0.22, 0.1, 0.0, 0.18, 0.05, 0.0]^T$$

$$\tilde{\mathbf{h}}_3^{(1)} = \mathbf{h}_3^{(0)} + \mathbf{a}_3^{(1)} = [0.45, 0.4, 0.52, 0.3, 0.25, 0.43, 0.35, 0.15]^T$$

9. Quantum Update Layer
A hybrid quantum circuit processes node embeddings.

Steps:
1. Classical projection (8 → 4 values)
2. Angle embedding using RY rotations
3. Variational quantum circuit with entanglement
4. Measurement of Pauli-Z expectation values
5. Classical projection back to embedding space

Quantum Update Layer

Let's use n_q = 4 qubits

StepA:Classical Pre-projection

Given:

$\tilde{\mathbf{h}}_3^{(1)} = [0.45, 0.4, 0.52, 0.3, 0.25, 0.43, 0.35, 0.15]^T$

W_in $\in \mathbb{R}^{4\text{x}8}$ = [[0.3, -0.2, 0.1, 0.4, -0.1, 0.2, 0.0, 0.3],
[-0.1, 0.4, 0.2, -0.3, 0.1, 0.0, 0.3, -0.2],
[0.2, 0.1, -0.4, 0.0, 0.3, -0.2, 0.1, 0.4],

[0.0, -0.3, 0.3, 0.2, -0.2, 0.4, -0.1, 0.0]]

b_in = [0.0, 0.1, -0.1, 0.05]

Compute:

z = W_in · h̃ + b_in

z[0] = 0.3(0.45) + (-0.2)(0.4) + 0.1(0.52) + 0.4(0.3) + (-0.1)(0.25)
+ 0.2(0.43) + 0.0(0.35) + 0.3(0.15) + 0.0
= 0.135 - 0.08 + 0.052 + 0.12 - 0.025 + 0.086 + 0 + 0.045
= 0.333

z = [0.333, 0.187, -0.142, 0.211]ᵀ

Apply tanh and scale by π:

$$\alpha = \pi \cdot \tanh(z)$$
$$= \pi \cdot [\tanh(0.333), \tanh(0.187), \tanh(-0.142), \tanh(0.211)]$$
$$= \pi \cdot [0.321, 0.185, -0.141, 0.208]$$
$$= [1.008, 0.581, -0.443, 0.653] \text{ radians}$$

$\alpha \in [-\pi, \pi]^4$ ✓

Step B: Angle Embedding

Each qubit starts in |0⟩ and is rotated by R_Y(α_q):

$$R\_Y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

For qubit 0 with α_0 = 1.008 rad:

$$R\_Y(1.008) = \begin{bmatrix} \cos(0.504) & -\sin(0.504) \\ \sin(0.504) & \cos(0.504) \end{bmatrix} = \begin{bmatrix} 0.876 & -0.482 \\ 0.482 & 0.876 \end{bmatrix}$$

$$|\psi_0\rangle = R\_Y(1.008)|0\rangle = \begin{bmatrix} 0.876 & -0.482 \\ 0.482 & 0.876 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.876 \\ 0.482 \end{bmatrix} = 0.876|0\rangle + 0.482|1\rangle$$

All qubit states after embedding:

$|\psi_0\rangle = 0.876|0\rangle + 0.482|1\rangle$
$|\psi_1\rangle = 0.958|0\rangle + 0.287|1\rangle$
$|\psi_2\rangle = 0.976|0\rangle - 0.220|1\rangle$
$|\psi_3\rangle = 0.947|0\rangle + 0.322|1\rangle$

Full 4-qubit initial state:

$|\psi\_init\rangle = |\psi_0\rangle \otimes |\psi_1\rangle \otimes |\psi_2\rangle \otimes |\psi_3\rangle \in \mathbb{C}^{16}$

For K = 2 variational layers, each layer has:

- 4 qubits × 3 angles = 12 parameters per layer
- Total: 24 trainable quantum parameters Θ

Single-qubit rotation

$U_q^{(k)} = R_Z(\theta_2) \cdot R_Y(\theta_1) \cdot R_Z(\theta_0)$

Example for qubit 0, layer 1:

$\theta_{1,0} = [0.5, 1.2, -0.3]$ (learned parameters)

$$R_Z(0.5) = \begin{bmatrix} e^{-i\cdot 0.25} & 0 \\ 0 & e^{i\cdot 0.25} \end{bmatrix}$$

$$R_Y(1.2) = \begin{bmatrix} \cos(0.6) & -\sin(0.6) \\ \sin(0.6) & \cos(0.6) \end{bmatrix}$$

$$R_Z(-0.3) = \begin{bmatrix} e^{i\cdot 0.15} & 0 \\ 0 & e^{-i\cdot 0.15} \end{bmatrix}$$

CNOT ring creates entanglement:

$CNOT_{0\to 1}$: flips qubit 1 if qubit 0 is $|1\rangle$
$CNOT_{1\to 2}$: flips qubit 2 if qubit 1 is $|1\rangle$
$CNOT_{2\to 3}$: flips qubit 3 if qubit 2 is $|1\rangle$
$CNOT_{3\to 0}$: flips qubit 0 if qubit 3 is $|1\rangle$ (completes the ring)

Step D: **Measurement**

After applying U(Θ), we measure the expectation value of Z on each qubit:

$o_q = \langle \psi\_final| Z_q |\psi\_final\rangle = P(|0\rangle) - P(|1\rangle) \in [-1, +1]$

Example measurement outcomes (simulated):
$o = [0.342, -0.156, 0.578, 0.089]^T$

Interpretation:

- $o_0 = 0.342$ means qubit 0 is measured as $|0\rangle$ ~67% of the time
- $o_1 = -0.156$ means qubit 1 is measured as $|1\rangle$ ~58% of the time

Step E: Classical Post-projection

$W\_out \in \mathbb{R}^{8x4}$

$q_3^{(1)} = W\_out \cdot o + b\_out$

$$
= \begin{bmatrix} 0.2 & -0.1 & 0.3 & 0.0 \\ -0.3 & 0.4 & 0.1 & 0.2 \\ 0.1 & 0.2 & -0.4 & 0.3 \\ 0.0 & -0.3 & 0.2 & -0.1 \\ 0.4 & 0.1 & 0.0 & 0.3 \\ -0.2 & 0.0 & 0.3 & 0.1 \\ 0.1 & 0.3 & -0.1 & 0.4 \\ 0.3 & -0.2 & 0.2 & -0.3 \end{bmatrix} \cdot \begin{bmatrix} 0.342 \\ -0.156 \\ 0.578 \\ 0.089 \end{bmatrix}
$$

$q_3^{(1)} = [0.232, -0.098, 0.145, 0.067, 0.178, 0.012, 0.089, 0.156]^T$

Step F: **Residual + LayerNorm**

$pre\_norm = \tilde{h}_3^{(1)} + q_3^{(1)}$

$= [0.45, 0.4, 0.52, 0.3, 0.25, 0.43, 0.35, 0.15]$
$+ [0.232, -0.098, 0.145, 0.067, 0.178, 0.012, 0.089, 0.156]$
$= [0.682, 0.302, 0.665, 0.367, 0.428, 0.442, 0.439, 0.306]$

LayerNorm calculation:

μ = mean(pre_norm) = 0.454
σ = std(pre_norm) = 0.138

$h_3^{(1)} = \gamma \odot (\text{pre\_norm} - \mu)/(\sigma + \varepsilon) + \beta$
With $\gamma = [1,1,1,1,1,1,1,1]$ and $\beta = [0,0,0,0,0,0,0,0]$:
$h_3^{(1)} = [1.65, -1.10, 1.53, -0.63, -0.19, -0.09, -0.11, -1.07]^T$

10. Layer Normalization
Final node representation:

Original (problematic):
h_i^(ℓ) = LN(h_i^(ℓ-1) + ∑... + Q_Θ(h_i^(ℓ-1) + ∑...))
This double-counts the aggregation term.
Corrected form:
h̃_i^(ℓ) = h_i^(ℓ-1) + ∑_{j∈N⁻(i)} MLP_msg([h_j^(ℓ-1) ‖ f_ij])
h_i^(ℓ) = LayerNorm(h̃_i^(ℓ) + Q_Θ^(ℓ)(h̃_i^(ℓ)))
Or equivalently:
h_i^(ℓ) = LayerNorm(h̃_i^(ℓ) + Q_Θ^(ℓ)(h̃_i^(ℓ)))
where h̃_i^(ℓ) = h_i^(ℓ-1) + ∑_{j∈N⁻(i)} MLP_msg([h_j^(ℓ-1) ‖ f_ij])

11. Edge-Level Traffic Prediction

Predicting traffic on edge ($v_1 \rightarrow v_2$):
h_1^(L) = [1.2, -0.5, 0.8, 0.3, -0.2, 1.1, 0.0, 0.6]^T
h_2^(L) = [0.9, 0.3, -0.4, 0.7, 0.5, -0.1, 0.8, 0.2]^T
f_{12} = [0.4, 0.1, 0.6, -0.3, 0.2, 0.5, -0.1, 0.3]^T

$r_{12} = [h_1\text{\^{}(L)} \parallel h_2\text{\^{}(L)} \parallel f_{12}] \in \mathbb{R}^{24}$

Prediction MLP:
W_pred $\in \mathbb{R}^{8\times24}$
hidden = ReLU(W_pred · r_{12} + b_pred) $\in \mathbb{R}^8$
w_out $\in \mathbb{R}^8$
$\hat{y}_{12}$ = w_out^T · hidden + b_out $\in \mathbb{R}$

Example output:
$\hat{y}_{12}$ = 0.73 (normalized traffic volume)

12. Denormalized Traffic Output
Convert normalized prediction to actual traffic volume:

To get actual vehicles/hour, denormalize:

$y\_actual = \hat{y}_{12} \cdot \sigma\_traffic + \mu\_traffic$
$= 0.73 \cdot 450 + 1200$
$= 1529$ veh/hr

13. Training Loss
Mean Squared Error

Loss Function & Optimization

6.1 MSE Loss Calculation

Ground truth for 5 training edges:
$y = [0.65, 0.82, 0.41, 0.93, 0.28]$
Predictions:
$\hat{y} = [0.73, 0.75, 0.45, 0.88, 0.31]$
Squared errors:
$(\hat{y} - y)^2 = [0.0064, 0.0049, 0.0016, 0.0025, 0.0009]$
$L = (1/5) \times (0.0064 + 0.0049 + 0.0016 + 0.0025 + 0.0009) = 0.00326$

14. Quantum Gradient Computation
Quantum parameters are trained using the Parameter-Shift Rule:

For quantum parameter θ affecting measurement $\langle Z\_q \rangle$:
$\partial\langle Z\_q \rangle/\partial\theta = (1/2)[\langle Z\_q \rangle(\theta + \pi/2) - \langle Z\_q \rangle(\theta - \pi/2)]$

Example:
Current $\theta_3 = 1.2$ rad
$\langle Z_0 \rangle$ at $\theta_3 = 1.2 + \pi/2 = 2.77$: forward_eval = 0.412
$\langle Z_0 \rangle$ at $\theta_3 = 1.2 - \pi/2 = -0.37$: backward_eval = -0.156
$\partial\langle Z_0 \rangle/\partial\theta_3 = (1/2)(0.412 - (-0.156)) = 0.284$

Cost: 2 circuit evaluations per quantum parameter per gradient computation.